

Experiment 6.3

Determination and plotting of Capacitance vs Dielectric Constant

Code File Name: Exp6_3_Capacitance_vs_DielectricConstant.m

Code

```
clc;
epsilon0 = 8.854e-12;
A = 1e-4;
d = 1e-3;
er = linspace(1,1.5,100);
C = (epsilon0 .* er .* A) ./ d;
plot(er,C,'LineWidth',2);
xlabel('Dielectric constant');
ylabel('Capacitance');
title('Capacitance vs dielectric constant (Air)');
```

Output Graph

